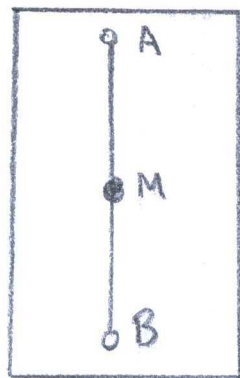


Euclidean and Non-Euclidean Geometry:

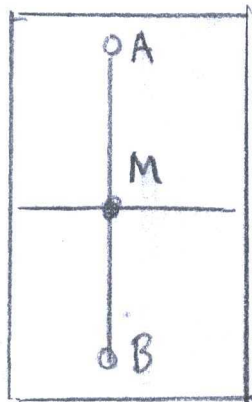
Exercises: a) Midpoint M of segment AB:

1.



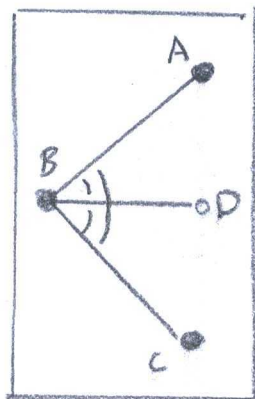
Definition: For a point M on segment $\overline{AB}$, a midpoint M exists when $\overline{AM} = \overline{MB}$.

b) Perpendicular bisector of segment AB:



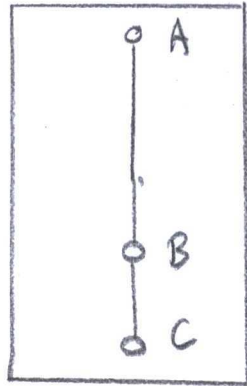
Definition: A line equal distance from both points A and B through midpoint M.

c) Ray BD bisects angle $\angle ABC$:



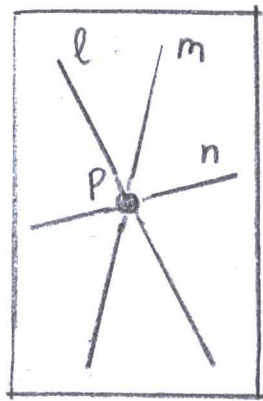
Definition: For an angle $\angle ABC$ and a point D in $\angle ABC$, the ray BD bisects when $\angle ABD = \angle DBC$.

d) Points A, B, C are collinear:



Definition: When lines AB and BC are on line AC.

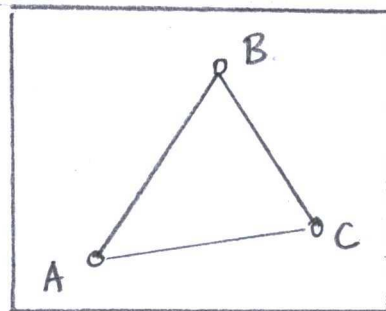
e) Lines l, m, and n are concurrent:



Definition: When points on lines l, m, and n are the same point.

2.

a) A triangle formed by three noncollinear points A, B, and C:

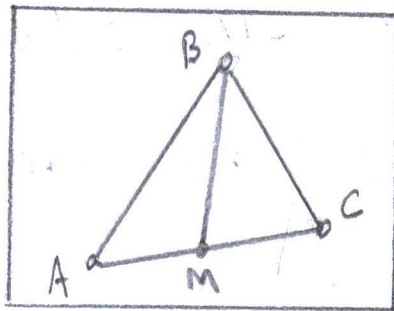


b) The vertices on triangle $\triangle ABC$ are points A, B, and C. Sides are segments AB, BC, and AC.

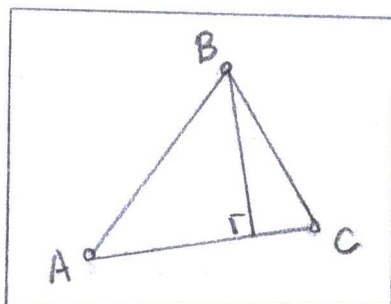
Angles on $\triangle ABC$ are $\angle ABC$, $\angle BCA$ and $\angle CBA$.

c) An adjacent side to vertex A is AB or AC. The third side is opposite, by deduction BC.

d) Medians on a triangle are bisectors from an opposite angle, such as $\overline{AC}$ to $\overline{AM} = \overline{MC}$.

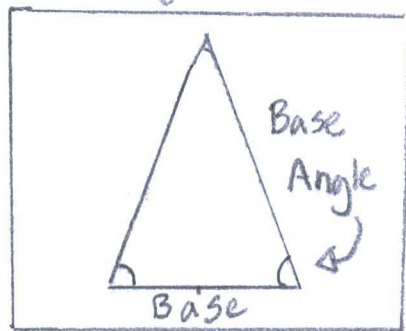


e) Altitudes on triangles are perpendiculars to a base from opposite angles.

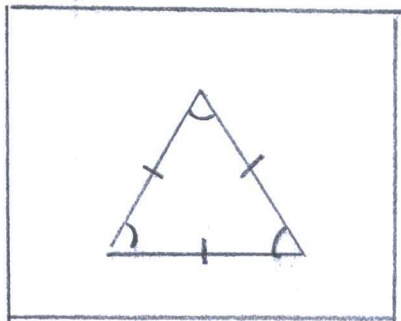


f) Isoscles triangle, its base, and

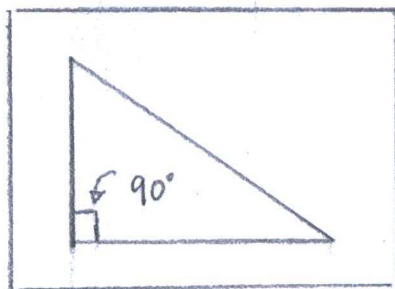
its base requires a different triangle. Two base angles are necessary to isosceles. Base segments have base angles.



g) Equilateral triangles have three equal sides and angles.

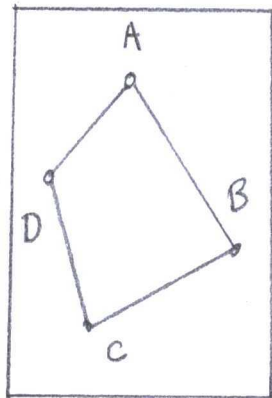


h) Right triangles have a 90° angle.



3.

a) The angles of $\square ABCD$:



Definition: The space between two segments at a vertex, such as $\angle A$ is the space between DA and AB.

b) Adjacent Sides of $\square ABCD$:

Definition: The segment intersecting another segment.

c) Opposite Sides of $\square ABCD$:

Definition: The side not intersecting a segment.

d) Diagonals of $\square ABCD$:

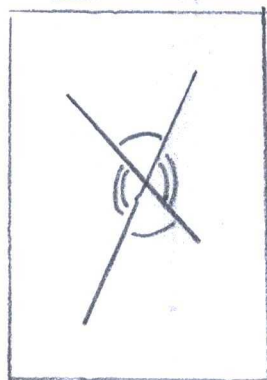
Definition: Segments connecting opposite vertices.

e) Parallelogram:

Definition: A quadrilateral having four vertices.

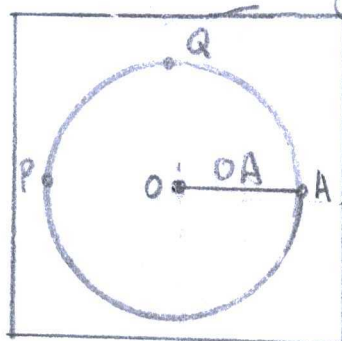
4.

Verticle Angles : Angles



Definition: Opposite angles across two intersecting lines.
Protractors measure angles.

5.

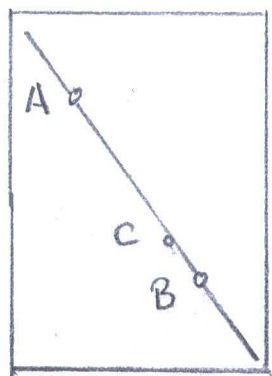


(pg 17) "Euclids Postulate III"

For every point O and every point A not equal to O, there exists a circle with center O and radius OA.

6.

a)



(pg 15) "Euclids Postulate I"

"For every point P and for every point Q not equal to P there exists a unique line that passes through P and Q."

b) $\overrightarrow{AB} = \overrightarrow{AC}$:

Proof

Hypothesis

Evidence

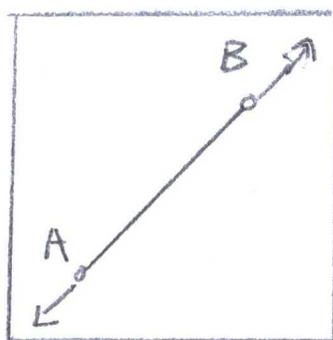
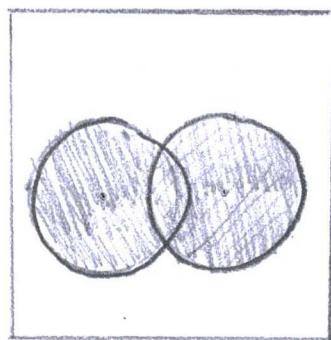
1) $\overrightarrow{AB} = \overrightarrow{AC}$

(pg 18) "Definition - Ray"

7.

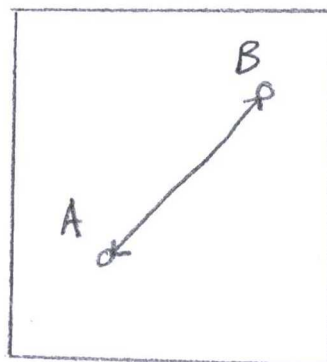
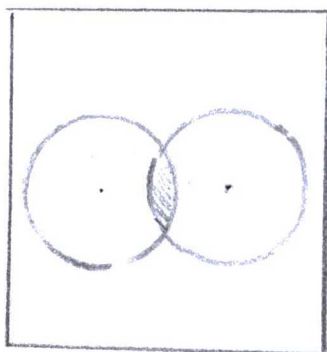
i) "Union"

$$\overleftrightarrow{AB} \cup \overleftrightarrow{BA} = \overleftrightarrow{AB}$$



ii) "Intersection"

$$\overleftrightarrow{AB} \cap \overleftrightarrow{BA} = \overline{AB}$$



(pg 18) "Definition - Ray"

Ray $\overrightarrow{AB}$ follows set of points lying on the line $\overleftrightarrow{AB}$: those points belong to segment $\overline{AB}$ and all points C on $\overleftrightarrow{AB}$,

Such B is between A and C .

The ray $\overrightarrow{AB}$ emanates from the vertex to be part of line $\overleftrightarrow{AB}$.

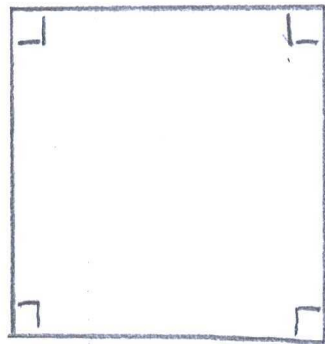
(pg 16) "Definition - Segment"

From distinct points A and B .

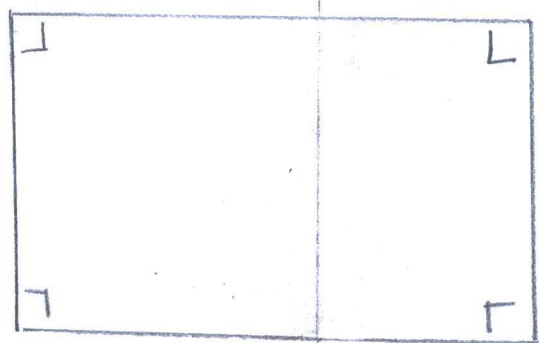
The segment AB is the set whose members are points A and B and all points C on line AB are between A and B . The two points A and B are endpoints of segment AB .

8.

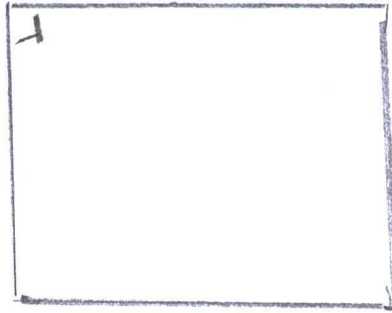
i) A quadrilateral with four right angles



ii) A quadrilateral with all angles congruent to one another:



iii) A parallelogram with at least one right angle:



9.

a) For every line exists a point on l :

Proof	<u>Hypothesis</u>	<u>Evidence</u>
	1) Line l has a point	Euclid Postulate I

b) For every line exists a point not on l :

Proof:	<u>Hypothesis</u>	<u>Evidence</u>
	1) Line l and m are parallel	Euclid's Parallel Postulate
	2) A point P lies on m	Euclid's Postulate I

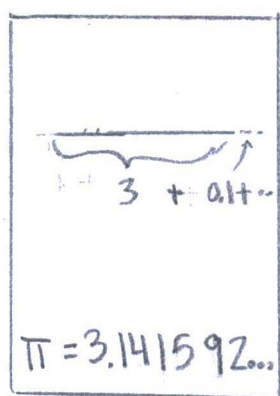
10.

A plane is non-empty because a set for all sets with elements for lines and points.

11. The Euclidean Parallel is obvious by modern day "computer generated images."

12. False fact all triangles are isosceles because the proof on page 25. Case 1-3 illustrate triangles not in the isosceles category.

13.



(from exercise) " π "

" π is the ratio of the circumference to its diameter."

A new definition to Φ , ratio of area in a circle to its diameter, has problems. Specifically, a major problem is Φ contains π .

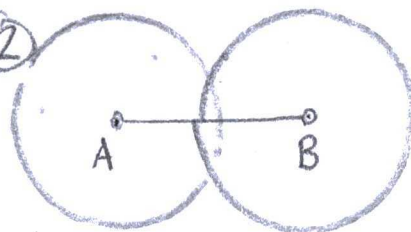
14.

a) Perpendicular bisector of AB:

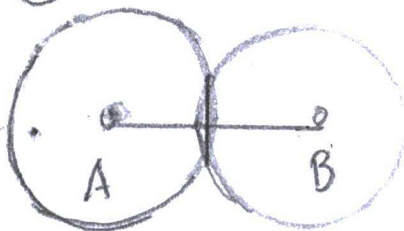
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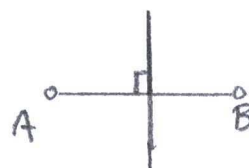
②



③



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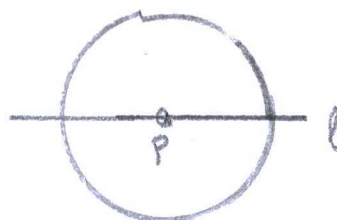


b) Line through P perpendicular to l .

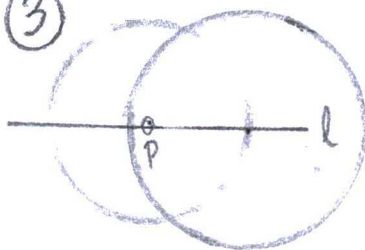
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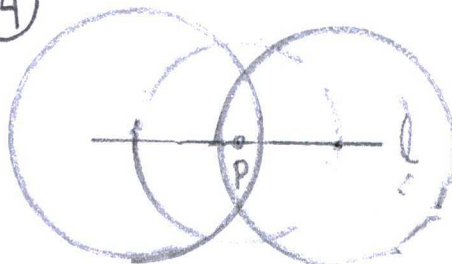
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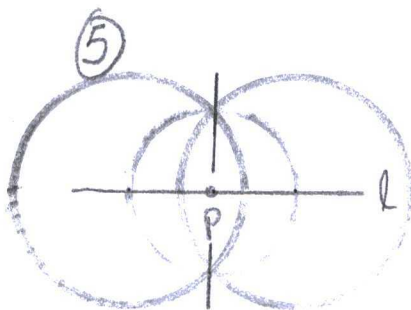
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④



⑤



⑥

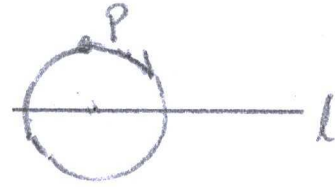


d) A line through P parallel to l:

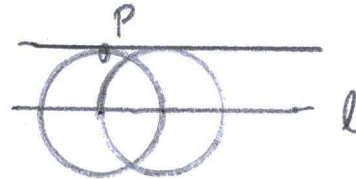
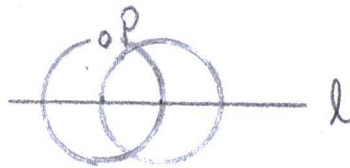
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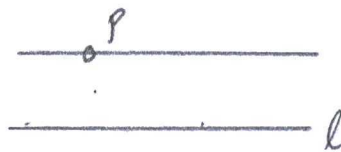
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③

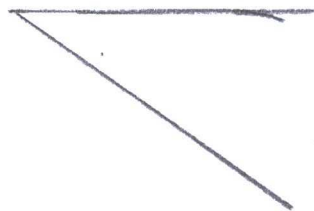


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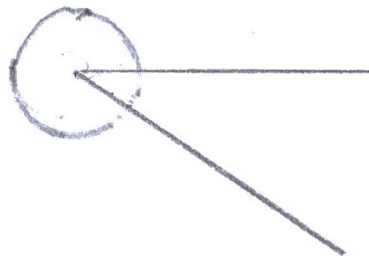


e) A bisecting ray of an angle:

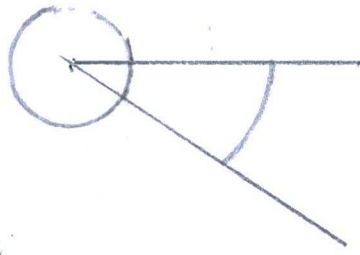
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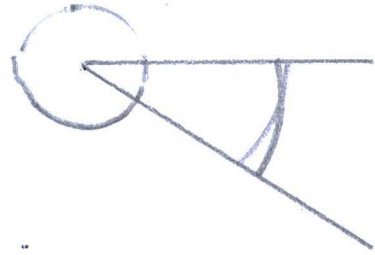
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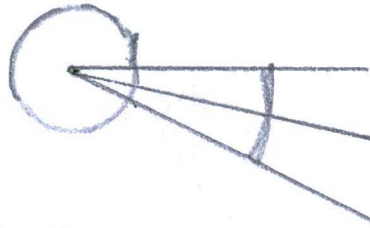
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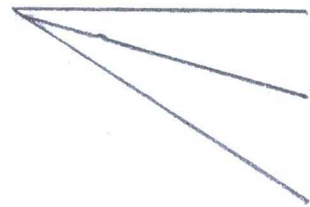
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⑤

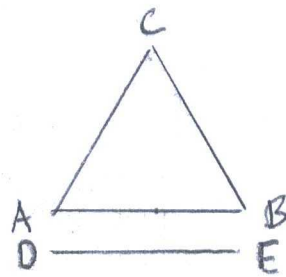


⑥

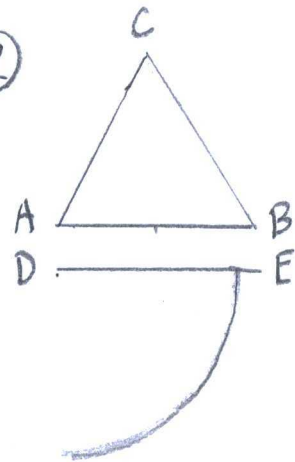


f) $\triangle DEF \cong \triangle ABC$:

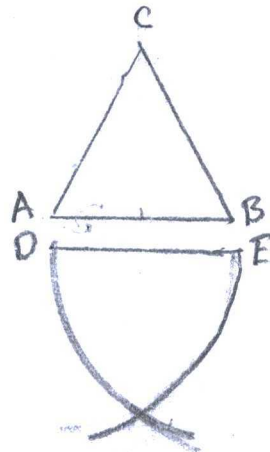
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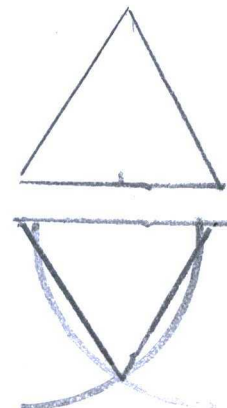
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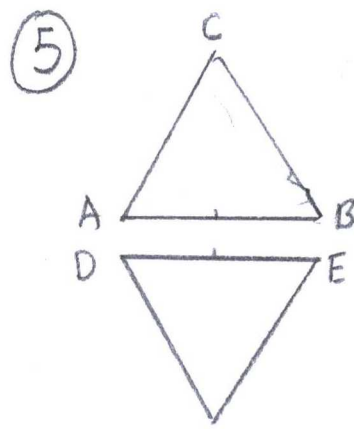


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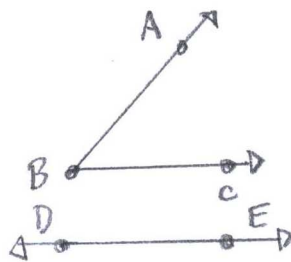
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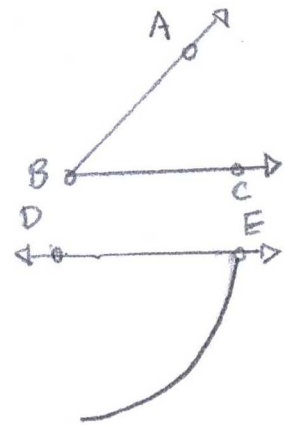


9) $\triangle ABC \cong \triangle FDE$

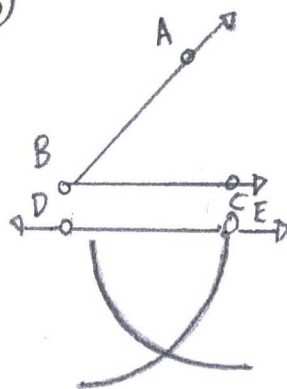
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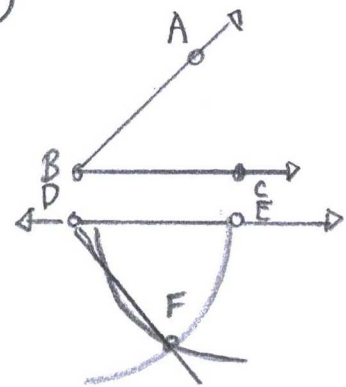
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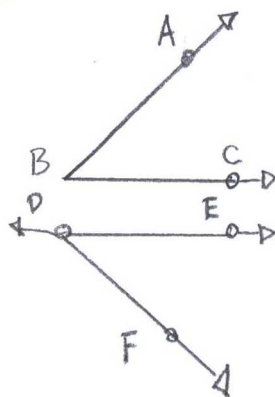
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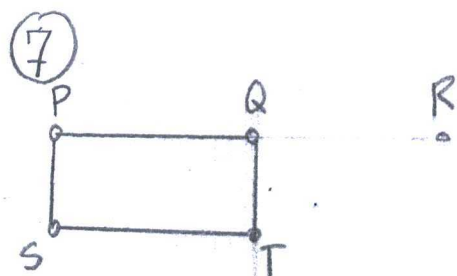
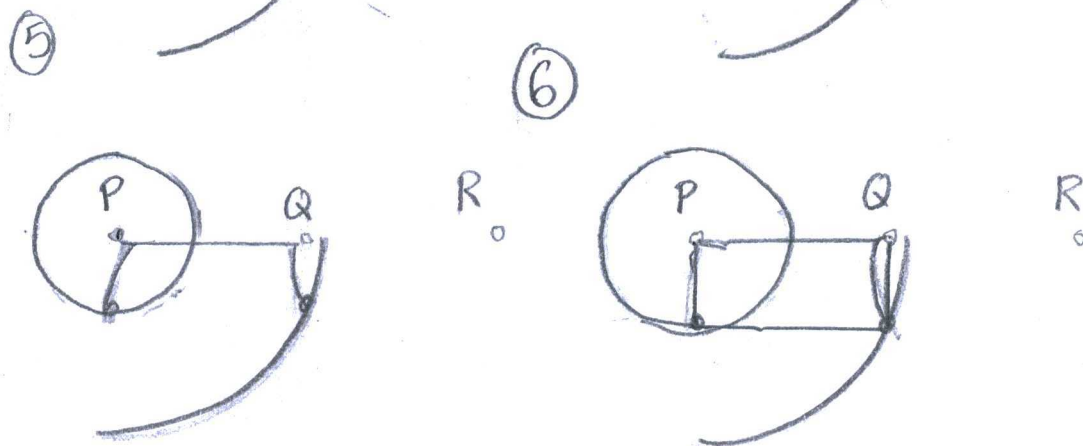
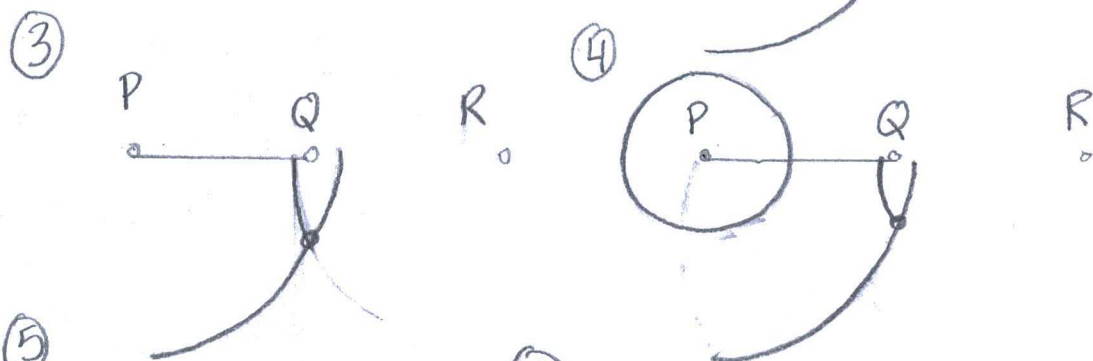


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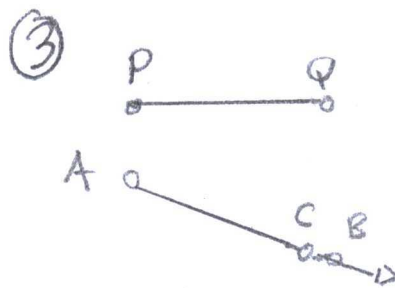
15.

a) Rectangle $\square PQST$ with $PT \cong PR$:

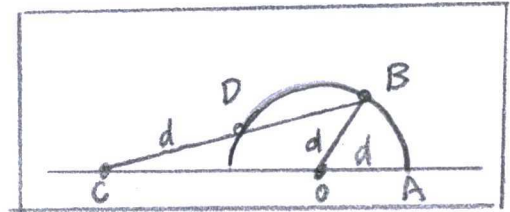
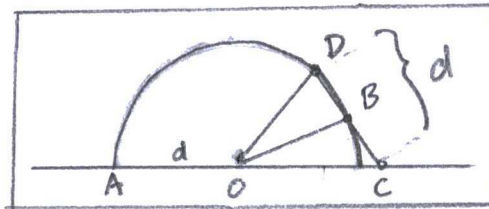


b) Point C on AB such that $PQ \cong AC$:





16.



$\angle COD$ is one-third $\angle AOB$:

Proof:	Hypothesis	Evidence
	1) If $\angle DCO \cong \angle DOC$	Conjecture
	2) $\angle AOB = \angle OCD + \angle OBD$	(pg 31) "I. 32"
	3) $\angle OBD \cong \angle ODB$ $\cong \angle OCD + \angle DOC$ $\cong 2\angle OCD$	(pg 31) "I. 5"
	4) $\angle AOB = \angle OCD + 2\angle OCD$	
	5) $\frac{1}{3} \angle AOB \cong \angle OCD$	Outcome